STAFF REVIEW · H446 311 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. A property true before and after each completed iteration.

2. Runs count contiguous groups; distinct counts globally unique values.

3. Otherwise a[0] could be invalid and initial count wrong.

4. Distinct values 2; runs 3 ([1,1],[2],[1]).

I Do — source worked steps

1. Predict first — Consecutive repeated fives belong to the same second run.

2. Explain and check — Explain the reason for the result.

We Do — guided CHECK

1. Predict first
There are four runs but only two distinct values.
Explain and check
Explain the reason for the result.

You Do — source mark schemes

1. Start runs1 [1]; change to4 makes2 [2]; change to3 makes3 [1]; change to5 makes4 [1]; return4 [1].

2. Mark scheme [8]: an int-returning method accepting int[] [1]; detect an empty array and return 0 [2]; initialise the run count to 1 [1]; loop from index 1 [1]; compare neighbouring values [1]; increment on change [1]; return the result [1]. Equivalent correct C# earns the same marks.
CODE:

```
static int CountRuns(int[] a)
{
if (a.Length == 0) return 0;
int runs = 1;
for (int i = 1; i < a.Length; i++)
{
if (a[i] != a[i - 1]) runs++;
}
return runs;
}
```

3. Runs3 [1], distinct2 [1]; adjacency explains difference [1]; O(n) time [2]; O(1) auxiliary storage [1].

Exit Ticket — source answer

1. False. A value can occur in more than one run.